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THE HONG KONG POLYTECHNIC UNIVERSITY

DEPARTMENT OF LAND SURVEYING AND GEO-INFORMATICS

Programme	**:**	34478 BSc (Hons) in Land Surveying and Geo-Informatics 34478-SY BSc (Hons) in Land Surveying and Geo-Informatics 34480-FYL BSc (Hons) in Land Surveying and Geo-Informatics 34480-SYL BSc (Hons) in Land Surveying and Geo-Informatics			
Subject Title	**:**	Geospatial Database Management and Design			
Subject Code	**:**	LSGI3245	**Session**	**:**	Semester 1, 2024/25
Date	**:**	12 December 2024	**Time**	**:**	12:30 – 14:30
Time Allowed	**:**	2 hours	**Subject Examiner(s)**	**:**	Dr YU Yue (LSGI)

This question paper has a total of 5 pages.
(Some pages may be intentionally omitted.)

Instructions to Candidates: This paper has **TWO** sections, A and B

Section A: Answer **ALL** questions.

Section B: Answer any **ONE** question.

Total marks = 100.

Available from Invigilator: Nil

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Section A: Answer **ALL** questions.

Question 1 (True/False)

Question 2

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Question 3

What is the definition of a spatial database? What is the definition of spatial data models? What is the definition of a spatial database management system (SDBMS)? (10 marks)

Question 4

What is the definition of an entity set in Entity-Relationship (ER) Model? What is the definition of relationship sets? List the types of mapping cardinality of a binary relationship set. (10 marks)

Question 5

List three different steps of database design. For each step, please use at most three sentences to describe the main functions or techniques applied. (10 marks)

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Question 6

A query language is a language in which a user requests information from the database. The relational algebra is a procedural query language which forms the theoretical basis of the SQL query language.
The relational algebra consists of a set of operations that take one or two relations as input and produce a new relation as their result. Please list the new relation using relational operations based on existing relation *r* and relation *s*:

(a) Union of two relations *r* and *s* ($r \cup s$) that returns a relation that includes all the tuples that appear in either or both *r* and *s*. (5 marks)

A	*B*
α	1
α	2
β	1

r

A	*B*
α	2
β	3

s

(b) Intersection of two relations *r* and *s* ($r \cap s$) that returns a relation that includes all the tuples that appear in both *r* and *s*. (5 marks)

A	*B*
α	1
α	2
β	1

r

A	*B*
α	2
β	3

s

(c) Set Difference of two relations *r* and *s* ($r - s$) that returns a relation that includes all the tuples that are contained in *r* but not in *s*. (5 marks)

A	*B*
α	1
α	2
β	1

r

A	*B*
α	2
β	3

s

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(d) Join Two Relations – Cartesian Product that combines tuples from two relations, and its result contains all pairs of tuples from the two relations, regardless of whether their attribute values match. (5 marks)

A	B
α	1
β	2

r

C	D	E
α	10	a
β	10	a
β	20	b
γ	10	b

s

- END OF SECTION A -

Section B: Answer any **ONE** question.

Question 7

List at least five points of drawbacks when using file systems. (10 marks)

Question 8

What is the definition of the storage manager and the main functions of the storage manager? (10 marks)

Question 9

A major purpose of a database system is to provide users with an abstract view of the data, hiding complexity from users. Please list different levels of abstraction and related definitions. (10 marks)

- END OF SECTION B -

- END OF PAPER -